Dictionary – Basic

Deutsch-Englisches Glossar statistischer Fachbegriffe

Grundbegriffe

#	Deutsch	English	Definition
1	population	**Population**	All elements/people about which a statement is to be made.
2	sample	**Sample**	Subset of the population that is actually being studied.
3	characteristic (variable)	variable	Property of an element that is measured.
4	characteristic expression	**Characteristic Value**	Specific value or category of a characteristic.
5	**Urliste**	**Raw Data**	Unsortierte Liste aller erhobenen Rohdaten.

Frequencies

#	Deutsch	English	Definition
1	Absolute frequency	**Absolute Frequency**	Number of occurrences of a characteristic expression.
2	Relative frequency	**Relative Frequency**	Absolute frequency divided by total number.
3	**Klassenbreite**	**Class Width**	Difference between upper and lower class limits.

Skalenniveaus

#	Deutsch	English	Definition
1	**Nominalskala**	**Nominal Scale**	Categories without order (e.g. gender, field of study).
2	**Ordinalskala**	**Ordinal Scale**	Values with natural order but without fixed spacing.
3	**Intervallskala**	**Interval Scale**	Equal distances, but no natural zero point (e.g. °C).
4	Ratio scale	**Ratio Scale**	Equal distances and absolute zero (e.g. size).

Lageparameter

#	Deutsch	English	Definition

1	**Arithmetisches Mittel**	**Mean**	Sum of all values divided by number.
2	**Median**	**Median**	Average value of a sorted data series. More robust than average.
3	**Modalwert (Modus)**	**Mode**	Most common value in a data series.

Streuungsparameter

#	Deutsch	English	Definition
1	**Spannweite (Range)**	**Range**	Difference between maximum and minimum.
2	Variance	**Variance**	Mean square deviation from the mean.
3	Standard deviation	**Standard Deviation**	Square root of the variance. Same unit as mean.

Wahrscheinlichkeit

#	Deutsch	English	Definition
1	**Zufallsexperiment**	**Random Experiment**	Process with uncertain outcome, repeatable.
2	**Ergebnisraum Ω**	**Sample Space Ω**	Set of all possible outcomes.
3	**Ereignis**	**Event**	Subset of the result space.
4	**Gegenereignis**	**Complement**	All results that are not in the event.
5	**Laplace-Wahrscheinlichkeit**	**Laplace Probability**	WS = (#favorable) / (#possible) for equal WS.
6	**Baumdiagramm**	**Tree Diagram**	Graphic of multi-stage random experiments with path rules.
7	**Pfadregel**	**Path Rule**	WS of a path = product of the individual WS.
8	**Bedingte Wahrscheinlichkeit**	**Conditional Probability**	$P(A \mid B) = P(A \cap B) / P(B)$.
9	**Vierfeldertafel**	**2×2 Contingency Table**	Table for conditional WS and stochastic independence.
10	Stochastic independence	**Stochastic Independence**	$P(A \cap B) = P(A) \cdot P(B)$.
11	**Zufallsvariable**	Random variables	Variables whose values depend on chance.
12	**Erwartungswert E(X)**	**Expected Value**	Average over infinite repetitions.
13	**Bernoulli-Experiment**	**Bernoulli Trial**	Two outcomes: success (p) and failure (1-p).

Verteilungen

#	Deutsch	English	Definition

1	**Binomialverteilung**	**Binomial Distribution**	Number of successes in n attempts with WS p.
2	**Normalverteilung**	**Normal Distribution**	Bell-shaped, symmetrical continuous distribution (μ, σ).
3	**Sigma-Regeln**	**Sigma Rules**	Approximately 68% (1σ), 95% (2σ), 99.7% (3σ) around μ at NV.

Testen

#	Deutsch	English	Definition
1	**Hypothesentest**	**Hypothesis Test**	Decision on H_0 based on sample.
2	**Nullhypothese H_0**	**Null Hypothesis H_0**	Annahme: kein Effekt / Unterschied.
3	**Alternativhypothese H_1**	**Alternative Hypothesis H_1**	Gegenthese: es gibt einen Effekt.
4	**Signifikanzniveau α**	**Significance Level α**	Max. WS for type 1 error. Usual: 0.05.
5	**Fehler 1. Art (α-Fehler)**	**Type I Error**	H_0 abgelehnt, obwohl wahr. Falsch-Positiv.
6	**Fehler 2. Art (β-Fehler)**	**Type II Error**	H_0 not rejected, although wrong. False negative.
7	**Ablehnungsbereich**	**Rejection Region**	Results where H_0 is rejected.
8	p-value	**p-value**	P(Result | H_0). $p < \alpha \rightarrow$ H_0 reject.